

HRFlux: An AI-Powered Multi-Agent HR Assistant.

Project Team

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Chapter 1

Introduction

This scope document is basically about setting the purpose, background, limits, and main outcomes for HRFlux, which is an AI-based HR automation system. The idea behind HRFlux is to cut down on repetitive manual work and make the overall employee experience smoother. In this document, I'll explain why the system is being created, what specific problems it aims to solve, the general technical approach, and also give an overview of the timeline and success measures for the project. HRFlux is built around multiple specialized agents that can chat with employees, a searchable HR knowledge base, automated handling of documents, reminders, and workflows for escalations. All these features come together to take care of routine HR tasks and provide employees with quick, consistent, and relevant answers. This proposal also connects back to the early research, prototypes, and requirements that were already shared in the project presentation and slides.

1.1 Problem Statement

HR teams often spend way too much time dealing with the same small questions over and over, like checking leave balances, providing NOCs, or explaining policies. For employees, finding clear answers is also a struggle because information is usually buried in long documents or hidden on intranet pages. This leads to repeated questions, unnecessary follow-ups, and slow approvals. On top of that, HR staff end up giving inconsistent responses, which adds to the confusion and floods their inboxes and chat channels.

The real problem is that all this manual work keeps HR people busy with routine tasks instead of letting them focus on bigger priorities like performance management, talent growth, or updating policies. Smaller HR teams feel this pressure even more, since even a small rise in routine queries can eat up a huge chunk of their time. Current chatbot or FAQ tools don't help much either—they usually give generic, out-of-context answers that frustrate employees and eventually push them to contact HR directly.

When it comes to generating documents, such as experience letters, NOCs, or approval memos, most of it is still done by hand, which is both slow and error-prone. There's also very little automation for reminders or escalations, so deadlines and SLAs often get missed. To make things worse, knowledge silos and poor search functions mean employees can't easily discover the benefits or entitlements available to them. Without session memory, the system can't handle multi-step or personalized interactions, which makes it less useful. Inconsistent answers also raise compliance and audit risks, especially when official approvals or documentation are involved.

HRFlux is being designed to tackle these problems head-on. The goal is to automate routine HR tasks, provide accurate and timely responses, and ease the workload on HR staff so they can concentrate on more meaningful, high-value work.

1.2 Motivation

The main drive behind this project is to make HR processes quicker, more reliable, and more focused on employees—without having to hire extra staff. By automating routine questions and workflows, we can cut down the time it takes to resolve issues, save costs, and also improve how satisfied employees feel with HR support. Using a modular, multi-agent setup means each agent can handle a specific area, which gives more accurate results than a generic chatbot. A searchable knowledge base also makes sure answers are always linked back to official policies instead of guesswork. On top of that, features like proactive reminders and escalation workflows help the company stick to SLAs and reduce mistakes that usually happen when everything is handled manually. Another big reason for doing this is to actually build a working prototype, which not only shows the real ROI of AI in HR but also helps the team gain hands-on experience in system integration, data privacy, and managing multiple agents working together.

1.3 Project Description

We are proposing **HRFlux**, an AI-powered HR assistant made up of multiple specialized agents. Instead of one generic chatbot, each agent has its own role—for example, *LeaveBot* for leave requests, *PolicyBot* for policy-related questions, and *DocumentBot* for generating letters or memos. Since every agent handles a specific area, the answers are more reliable and easier to track. These agents work together through a simple coordination layer so they can carry out step-by-step tasks like applying for leave, routing approvals, or creating official documents.

To answer policy questions, the system uses retrieval-augmented methods that combine

semantic search with extractive QA, ensuring that responses are grounded in the organization's actual HR documents. A memory component also keeps track of the context across sessions, so employees don't have to repeat the same details every time, making interactions feel more personal.

For documents, the system uses predefined templates (such as NOCs, experience letters, or approval forms) to save time and reduce errors, while keeping everything consistent and compliant. HRFlux also includes proactive reminders and notifications to prevent missed deadlines or SLA breaches. If a case cannot be resolved automatically, it is escalated to the right HR officer along with the full conversation history, ensuring smooth handover and accountability.

The system has two main parts: a chatbot interface for employees and an admin dashboard for HR staff, which lets them review, approve, or adjust outputs when needed. Built-in safeguards like role-based access, privacy controls, and audit logs keep it compliant with organizational standards.

To integrate smoothly into existing systems, HRFlux will connect with HRIS and payroll platforms through APIs, allowing updates like leave status to be managed automatically. The system will also track performance through metrics such as response time, resolution rate, number of escalations, and hours saved for HR staff.

Development will follow an iterative approach, starting with the core knowledge base and main agents, then gradually adding features like document automation, reminders, memory, and admin tools. The end goal is to deliver a working prototype that reduces HR workload, improves consistency, and demonstrates real productivity gains.

1.4 Project Objectives

- Automate answers to the top 80% of routine HR queries (leave, NOC, policy clarifications).
- Implement specialized agents (*LeaveBot*, *PolicyBot*, *DocumentBot*) with domain accuracy.
- Build a semantic, searchable HR knowledge base that ingests policy documents and FAQs.
- Add document automation templates for NOCs, experience letters, and approval memos.
- Implement contextual session memory for personalized multi-step interactions.
- Provide configurable proactive reminders and escalation workflows.

- Deliver an HR admin dashboard for auditing, overrides, and metrics.
- Integrate with at least one HRIS/payroll system via API for read/write of leave data.
- Ensure role-based access control, logging, and privacy compliance.
- Measure and report KPIs: average resolution time, resolution rate, escalations, and estimated HR hours saved.

1.5 Stake Holders

- HR managers and HR officers (primary users and approvers).
- Employees (end users who will ask queries and receive documents).
- IT/DevOps (integration, deployment, and API connectivity).
- Legal Compliance (review policies, privacy, audit requirements).

Chapter 2

Project Description

2.1 Scope

The proposed system, HRFlux, is designed in a modular way to ensure clarity, scalability, and efficient development. Each module is responsible for a specific set of functions, and together they achieve the overall objectives of the system. The backend modules handle data management, knowledge processing, and AI-driven agent operations. The agent layer is responsible for query handling, document generation, and task escalation. On the front end, employees interact with the system through a chatbot, while HR staff use an admin portal for management and approvals. This separation ensures that both employees and HR teams have tailored interfaces suited to their needs. The modular design also allows for independent development and testing, making the system more robust. Overall, these modules work together to reduce repetitive HR tasks, improve response times, and streamline operations across the organization.

2.2 Modules

The proposed system, HRFlux, is designed in a modular way to ensure clarity, scalability, and efficient development. Each module is responsible for a specific set of functions, and together they achieve the overall objectives of the system. The backend modules handle data management, knowledge processing, and AI-driven agent operations. The agent layer is responsible for query handling, document generation, and task escalation.

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2.2.1 Module 1: Data Collection & Database Management

This module is responsible for collecting, organizing, and storing HR-related data. It ensures centralized access to employee records, leave balances, and policy documents, forming the foundation of the system.

1. Employee Database – Stores employee profiles, leave balances, and payroll details.
2. Policy Document Upload – Allows HR staff to upload and manage company policies.
3. Structured HR Data – Integrates mock HRMS/Payroll data for workflows.
4. Knowledge Preparation – Prepares documents for search and retrieval.

2.2.2 Module 2: Knowledge Base & Backend Services

This module transforms stored data into a usable knowledge base and provides backend APIs for agents. It supports semantic search, workflow management, and system integrations.

1. Vector Database Integration – Embeds and indexes HR documents for semantic search.
2. Backend APIs – Provide access to employee data, leave balances, and payroll info.
3. Workflow Engine – Handles leave requests, approvals, and rejections.
4. Search & Retrieval – Delivers accurate responses from policy documents.

2.2.3 Module 3: HR Agent Layer

This module introduces specialized AI-driven agents, each focused on a specific HR function. These agents work with backend services to resolve employee queries and generate documents.

1. LeaveBot – Handles leave balance queries and applications.
2. PolicyBot – Responds to HR policy and guideline questions.

3. DocuBot – Generates HR documents such as NOC letters and leave approvals.
4. EscalationBot – Routes unresolved or sensitive queries to HR staff.

2.2.4 Module 4: Interfaces (Employee & Admin Portals)

This module provides interaction points for employees and HR staff. Employees use a chatbot for queries, while HR staff access an admin portal for management and approvals.

1. Employee Chatbot – Allows leave applications, document requests, and policy checks.
2. Admin Portal – Enables HR staff to manage profiles, policies, and leave approvals.
3. Query Logs – Maintains records of employee queries for monitoring.
4. Approval Dashboard – Provides tools for HR staff to handle approvals and escalations.

2.3 Tools and Technologies

The development of HRFlux requires a combination of tools and technologies that cover data storage, backend services, AI agent development, and user interface design. These tools ensure smooth integration between modules and provide a scalable environment for future enhancements. The major tools and technologies are as follows:

1. **Programming Languages:** Python for implementing AI agents, backend services, and integration; JavaScript/TypeScript for frontend development.
2. **Frameworks:** FastAPI for building backend APIs and services; React.js for developing the employee chatbot interface and admin web portal.
3. **Databases:** MySQL or PostgreSQL for structured employee and HR data; FAISS or Chroma for vector database to enable semantic search of policy documents.
4. **AI Libraries:** LangChain, AutoGen, or LangGraph for creating and managing multi-agent workflows; Hugging Face Transformers for NLP tasks.
5. **Document Processing:** PyPDF2, PDFplumber, or related libraries for extracting and embedding HR policy documents.
6. **Frontend Tools:** HTML, CSS, and Bootstrap/Tailwind for building responsive and user-friendly interfaces.

7. **Version Control:** Git and GitHub for collaborative development and version management.
8. **Deployment Tools:** Docker for containerization and easy deployment of backend services; WebSockets for enabling real-time communication between chatbot and backend.
9. **Collaboration Tools:** Slack or Microsoft Teams integration for cross-platform query handling; Postman for API testing.

2.4 Work Division

The tasks of the project are divided equally among the three team members. Each member is assigned features from all modules to ensure balanced participation and collective ownership of the system.

Table 2.1: Table 1: Work Division

Name	Registration	Responsibility / Module / Feature
Ms. Fatima	22i-0475	(Module 1 – Feat 1-2) Data collection and database design. (Module 2 – Feat 1) Policy document upload. (Module 3 – Feat 2) EscalationBot development. (Module 4 – Feat 1) Employee chatbot
Ms. Shayane	22i-1049	(Module 1 – Feat 3) Data preprocessing. (Module 2 – Feat 2) Knowledge base embedding and retrieval. (Module 3 – Feat 1,3) LeaveBot and PolicyBot development. (Module 4 – Feat 2) Query logs and tracking
Ms. Hadia	22i-087	(Module 1 – Feat 4) HR policy document integration. (Module 1 – Feat 5) Policy search and answering. (Module 3 – Feat 4) DocuBot for auto-document generation. (Module 4 – Feat 3) Admin dashboard and employee profile management

2.5 Timeline

The project timeline is divided into four major iterations. Each iteration focuses on a distinct set of tasks, ensuring progressive development and integration of all system com-

ponents.

Table 2.2: Table 2: Project Timeline

Iteration #	Time Frame	Tasks
01	Sept–Oct	Data collection, database design, and storage setup
02	Nov–Dec	Document preprocessing, policy retrieval, and embedding generation
03	Jan–Feb	Multi-agent workflow development and query handling
04	Mar–Apr	Admin dashboard creation, integration, and testing

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